SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1506

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I ____N.NITHIYASHREE-192421101_________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _______M.NITHIYASHREE_____________

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ANSWERS

1. Media Company Video Streaming Platform Migration

Scenario

A media company plans to migrate its video streaming platform to the cloud. The platform must support millions of users simultaneously while providing uninterrupted video playback with minimal buffering.

Cloud Architecture

- Deploy application servers in multiple Availability Zones (AZs).
- Use a **Content Delivery Network (CDN)** to cache video content closer to users.
- Store videos in cloud object storage (e.g., Amazon S3, Azure Blob Storage).
- Use a **Load Balancer** to distribute user requests across multiple servers.
- Enable **Auto Scaling** to automatically add or remove servers based on demand.
- Use a managed database service with read replicas for metadata.

Load Balancing and Auto-Scaling

- **Load Balancer:** evenly distributes traffic among servers to prevent overload.
- **Auto Scaling:** Adds new servers during peak demand and removes them during low traffic, ensuring smooth streaming and reducing buffering.

Cost Optimization Strategies

- Use auto-scaling to avoid paying for idle resources.
- Cache frequently accessed videos using CDN to reduce bandwidth costs.
- Store rarely accessed videos in low-cost archive storage.
- Use Reserved or Spot Instances for non-critical workloads.
- Monitor resource usage and optimize unused resources.

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2. Fintech Startup Using Containers Across Multiple Clouds

Scenario

A fintech startup deploys containerized applications across AWS, Azure, and Google Cloud. The company faces deployment consistency and management challenges.

Proposed Solution

- Use **Kubernetes** for container orchestration.
- Package applications using **Docker** containers.
- Use Infrastructure as Code (Terraform) for consistent deployment across cloud providers.
- Use a centralized CI/CD pipeline for automated deployments.

Service Discovery and Scaling

- Kubernetes Service Discovery automatically locates running services.
- Use Kubernetes DNS for communication between containers.
- Enable **Horizontal Pod Autoscaler (HPA)** to scale containers based on CPU or memory usage.
- Use Cluster Autoscaler to add or remove worker nodes automatically.

Benefits

- High availability.
- Vendor independence.
- Better disaster recovery.
- Improved scalability.
- Flexibility to choose cloud services.

Risks

- Increased management complexity.
- Different cloud security policies.
- Higher networking costs.
- Data synchronization challenges.
- Monitoring across multiple providers is more difficult.

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3. E-Learning Platform Backup and Disaster Recovery

Scenario

An e-learning platform loses important data due to accidental deletion. The organization requires a reliable backup and disaster recovery solution.

Proposed Solution

- Enable automatic cloud-native backups.
- Enable object storage versioning to recover deleted files.
- Replicate data to another cloud region.
- Schedule regular database snapshots.
- Store backups in geographically separate locations.

Achieving RTO and RPO

Recovery Time Objective (RTO):

- Use standby servers and automated failover to restore services quickly.
- Keep recovery procedures automated to reduce downtime.

Recovery Point Objective (RPO):

- Perform frequent backups.
- Enable continuous data replication to minimize data loss.

Business Continuity Steps

1. Enable automatic daily backups.
2. Replicate critical data across regions.
3. Test disaster recovery regularly.
4. Automate failover to backup systems.
5. Monitor backup status continuously.
6. Document and update disaster recovery procedures.

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4. Government Agency Hybrid Cloud

Scenario

A government agency stores sensitive citizen data on-premises while hosting public applications in the cloud due to regulatory compliance requirements.

Hybrid Cloud Architecture

- Keep confidential databases and critical applications in the on-premises data center.
- Host public websites and non-sensitive applications in the public cloud.
- Connect both environments using a secure VPN or dedicated private connection.

- Use firewalls and network segmentation for secure communication.

Identity Management

- Implement Single Sign-On (SSO).
- Use Identity Federation with SAML or OpenID Connect.
- Enable Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Store encryption keys securely using a Key Management Service (KMS).

Challenges

- Complex infrastructure management.
- Compliance with government regulations.
- Latency between cloud and on-premises systems.
- Security policy consistency.
- Higher operational complexity and cost.

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5. SaaS Provider Multi-Region High Availability

Scenario

A SaaS provider requires continuous service availability even if an entire cloud region becomes unavailable.

Multi-Region Deployment Strategy

- Deploy application servers in two or more cloud regions.
- Use a Global Load Balancer to route users to the nearest healthy region.
- Store application data in managed databases with cross-region replication.
- Maintain standby infrastructure in secondary regions.

Data Replication and Failover

- Enable synchronous replication within the same region for high consistency.

- Use asynchronous replication between regions for disaster recovery.
- Configure automatic failover to redirect traffic if the primary region fails.
- Regularly test failover procedures.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network usage.
- Track application response time and availability.
- Configure alerts for server failures, high latency, and unusual traffic.
- Use centralized logging and dashboards for real-time monitoring.
- Perform regular health checks and automated incident notifications.

Conclusion

Cloud technologies provide scalable, secure, and reliable solutions for modern applications. Using **CDNs, load balancing, and auto-scaling** ensures smooth media streaming. **Kubernetes and multi-cloud strategies** simplify container deployment. **Cloud-native backup, replication, and disaster recovery** protect data and reduce downtime. **Hybrid cloud architectures** balance security and compliance, while **multi-region deployments** provide high availability and fault tolerance. These approaches improve performance, resilience, and cost efficiency while ensuring business continuity.